

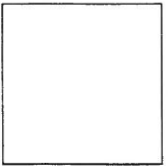
Surname / Van: Initials / Voorletters:

Student number / Studentenommer:

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MEMO



FSK 116

Semester Test 2 - Semestertoets 2

Department of Physics, University of Pretoria
Departement Fisika, Universiteit van Pretoria

Date/Datum: 11/05/2017 **Time/Tyd:** 2.5 hours/ure **Total (max.) / Totaal (maks.):** 70

Examiner/Eksaminator: RQ Odendaal **Moderator/Moderator:** Dr JM Nel

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Test register: Tear carefully on the line above and hand to the invigilator.
Toets register: Skeur versigtig op die lyn hierbo en handig dit in by die toesighouer.

FSK 116: Semester Test 2 / Semestertoets 2

11/05/2017

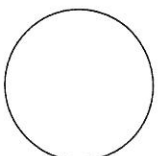
Surname <i>Van</i>		Initials <i>Voorletters</i>	
Student number <i>Studentenommer</i>			
Venue <i>Lokaal</i>			
Signature <i>Handtekening</i>			

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1. Answer all the questions in the space provided on the question paper. *Antwoord alle vrae in die spasie gelaat op die vraestel.*
2. No pencil work will be marked. Do not use "tippex". *Geen potlood-werk sal gemerk word nie. Moet nie uitwis-vloeistof ("tippex") gebruik nie.*
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues. *As u meer spasie benodig gebruik die agterkant van die bladsye. Dui duidelik aan waar die vraag aangaan.*
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams). *Teken diagramme waar nodig (bv. vryliggaams-diagramme).*
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.** *U antwoorde moet sistematies en logies neergeskryf word. U moet bewys dat u die nodige fisika verstaan en kan toepas. Numeriese waardes mag slegs in die laaste stap van 'n berekening vervang word! 'n Korrekte antwoord beteken nie dat u noodwendig vol punte sal kry nie.*
6. **Units must be used when numbers are substituted else marks will be subtracted.** *Eenhede moet gebruik word wanneer numeriese waardes vervang word, andersins sal punte afgetrek word.*
7. The use of the cheat sheet is only allowed subject to the rules stated on it. *Die formuleblad mag slegs gebruik word onderhewig aan die reëls daarop gedruk.*
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner. *Alle eksamen regulasies van die Universiteit van Pretoria geld en enige oortreding daarvan sal in 'n ernstige lig beskou word.*

Constants and formulas / Konstantes en formules

$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$; $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$; $N_A = 6.02 \times 10^{23} / \text{mol}$; $1 \text{ atm} = 101.3 \text{ kPa}$; $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 3 \times 10^8 \text{ m/s}$; $V_{\text{sphere/sfeer}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$; $1 \text{ litre} = 1000 \text{ cm}^3$; $M_{\text{Earth}} = M_{\text{Aarde}} = 5.97 \times 10^{24} \text{ kg}$, $\rho(\text{water}) = 1 \text{ g/cm}^3$; $r_{\text{Earth}} = r_{\text{Aarde}} = 6.37 \times 10^6 \text{ m}$

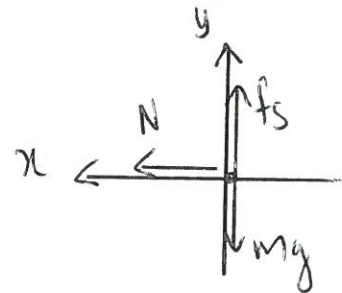
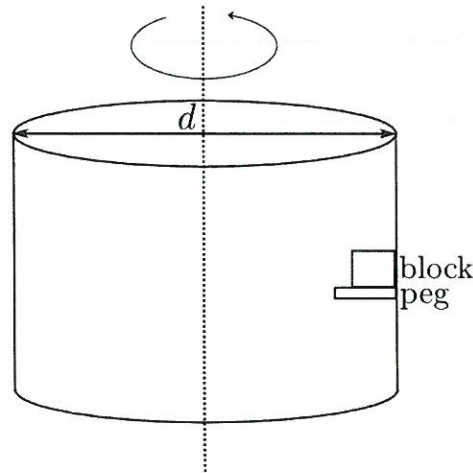


1 Question 1 / Vraag 1

[7]

A small block of mass m is in contact with the inner wall of a large hollow cylinder (diameter d). Assume the coefficient of static friction between the block and the wall of the cylinder is μ_s . Initially, the cylinder is at rest, and the block is held in place by a peg supporting its weight. The cylinder starts rotating about its center axis, as shown in the figure, with an angular acceleration α . Determine the minimum time interval (in terms of d , μ_s , g and α) after the cylinder begins to rotate before the peg can be removed without the block sliding against the wall.

'n Klein blokkie met massa m is in kontak met die binnewand van 'n hol silinder (diameter d). Neem aan die koëffisiënt van statiese wrywing tussen die blokkie en die wand van die silinder is μ_s . Die silinder is aanvanklik in rus, en die blokkie word in plek gehou deur 'n stokkie wat die gewig van die blokkie ondersteun. Die silinder begin roteer om die as deur die middel, soos getoon in die figuur, met 'n hoekversnelling α . Bepaal die minimum tydsinterval (in terme van d , μ_s , g en α) nadat die silinder begin roteer voordat die stokkie verwyder kan word sonder dat die blokkie teen die wand kan afgly.



$$\sum F_y = 0$$

$$f_s - mg = 0$$

$$f_s = mg$$

$$\mu_s N = mg \quad (1)$$

$$\sum F_x = \text{max} = m \frac{v^2}{r}$$

$$N = \frac{mv^2}{r}$$

$$\text{but } r = \frac{d}{2} \text{ and } v = \omega r$$

$$\Rightarrow N = \frac{m\omega^2 r^2}{r} = m\omega^2 r$$

$$\Rightarrow N = m\omega^2 \left(\frac{d}{2}\right) \quad (2)$$

Substitute (2) into (1):

$$\mu_s m\omega^2 \left(\frac{d}{2}\right) = mg \quad \text{but } \omega = \alpha t$$

$$\Rightarrow \mu_s \alpha^2 t^2 \left(\frac{d}{2}\right) = g$$

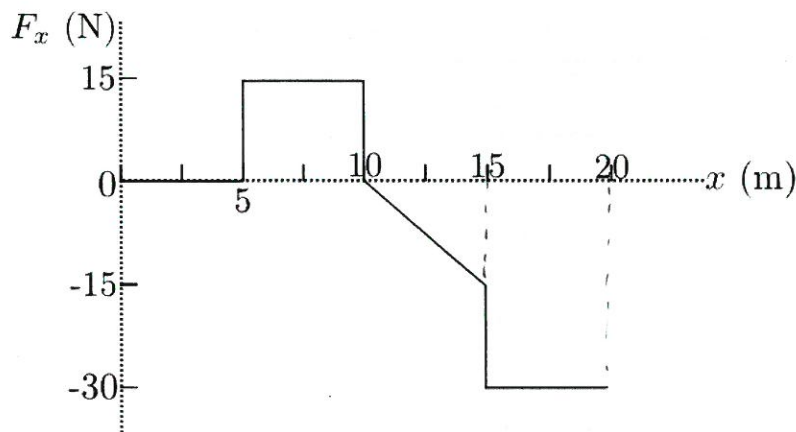
$$t_{\min} = \sqrt{\frac{2g}{\mu_s \alpha^2}} = \sqrt{\frac{2g}{\mu_s \alpha}} \cdot \frac{1}{\alpha}$$

2 Question 2 / Vraag 2

[5]

A varying force is exerted on a particle as it moves in the positive direction along the x -axis. The figure shows a graph of this force F_x versus the particle's position x . Find the work done by this force on the particle as the particle moves from $x_i = 0$ to $x_f = 20$ m.

'n Variëerbare krag word uitgeoefen op 'n deeltjie soos dit in die positiewe rigting langs die x -as beweeg. Die figuur toon 'n grafiek van hierdie krag F_x teenoor die deeltjie se posisie x . Vind die arbeid verrig deur hierdie krag op die deeltjie, soos hierdie deeltjie vanaf $x_i = 0$ tot by $x_f = 20$ m beweeg.



W_1 between $x=0$ and $x=5$ m is zero $\Rightarrow W_1 = 0$

$W_2 = (15\text{ N})(5\text{ m}) = +75\text{ J}$ between $x=5$ m and $x=10$ m

$W_3 = -\frac{1}{2}(5\text{ m})(-15\text{ N}) = -37.5\text{ J}$ between $x=10$ m and $x=15$ m

$W_4 = -(30\text{ N})(5\text{ m}) = -150\text{ J}$ between $x=15$ m and $x=20$ m

Net work done between $x_i=0$ and $x_f=20$ m is

$$W_{\text{net}} = W_1 + W_2 + W_3 + W_4$$

$$= 0 + 75\text{ J} - 37.5\text{ J} - 150\text{ J} = -112.5\text{ J}$$

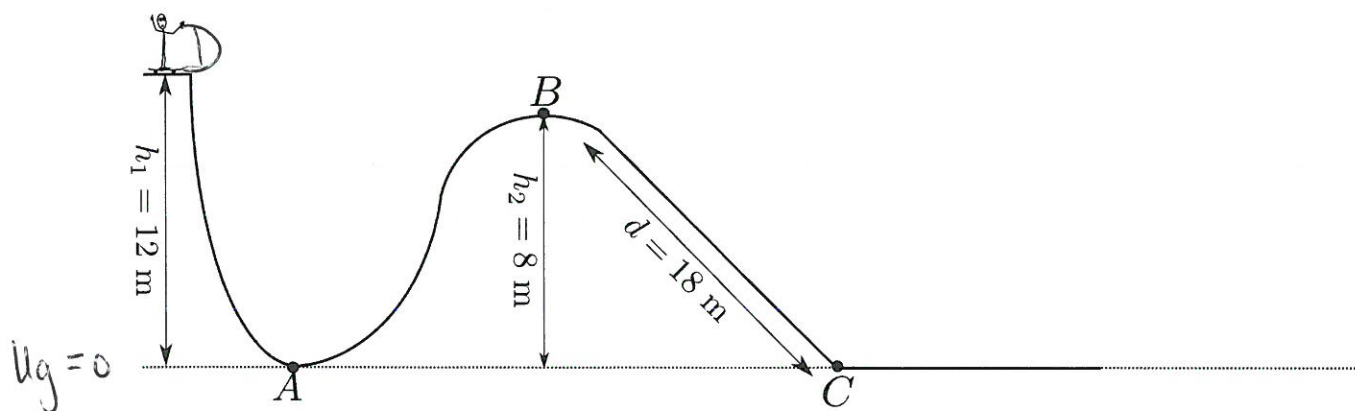
check sign!

3 Question 3 / Vraag 3

[12]

A boy ($m = 45$ kg) rides his skateboard at a skate park. He starts from rest at the top of the track as seen in the figure (not drawn to scale) and begins to descend down the track, always maintaining contact with the track surface. The mass of the skateboard is negligible, as is friction except where noted. In each question, start with the conservation of energy equation $\Delta K + \Delta U + \Delta E_{\text{int}} = W_{\text{external forces}}$ and clearly and briefly explain **your choice of initial and final values** for each quantity in this equation that is applicable to the situation described.

'n Seun ($m = 45$ kg) ry op sy skaatsplank by 'n oefenbaan. Hy begin aan die bopunt van die baan soos aangetoon in die figuur (nie volgens skaal geteken nie) en beweeg teen die baan af. Hy behou kontak met die baan-oppervlak ten alle tye. Die massa van die skaatsplank is weglaatbaar klein, sowel as wrywing behalwe waar aangedui. Vir elke vraag, begin met die behoud van energie vergelyking $\Delta K + \Delta U + \Delta E_{\text{int}} = W_{\text{external forces}}$ en verduidelik kortliks **jou keuses van aanvangs- en finale waardes** vir elke hoeveelheid in die vergelyking, vir elke situasie.



- (a) What is his speed when he reaches the bottom of the initial dip (point A)?

Wat is sy speed wanneer hy die onderpunt van die aanvanklike put (punt A) bereik?

(4)

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W_{\text{ext}} ; \quad \Delta E_{\text{int}} = 0 \quad \checkmark \text{ no friction}$$

$$\Rightarrow \Delta K + \Delta U = 0$$

$$K_f - K_i = -(U_f - U_i)$$

$$\frac{1}{2} m v_A^2 - 0 = -(0 - mgh_1)$$

$$\frac{1}{2} m v_A^2 = +mgh_1$$

$$v_A = \sqrt{2gh_1} = \sqrt{2(9.8 \text{ m/s}^2)(12 \text{ m})}$$

$$v_A = 15.3 \text{ m/s} \quad \checkmark$$

forces, $W_{\text{ext}} = 0$
 isolated system
 final point is A
 initial point is at D

- (b) He then ascends the other side of the dip to the top of a hill, 8.0 m above the ground. What is his speed at point B?

Hy beweeg opwaarts aan die anderkant van die put tot by die bopunt van die heuwel, 8.0 m bokant die grond. Wat is sy speed by punt B?

(4)

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W_{\text{ext}} ; \quad \Delta E_{\text{int}} = 0 \quad \checkmark \text{ no friction}$$

$$\Rightarrow \Delta K + \Delta U = 0$$

$W_{\text{ext}} = 0$
 isolated system

$$K_f - K_i = -(U_f - U_i)$$

$$\frac{1}{2} m v_B^2 - \frac{1}{2} m v_A^2 = -(mgh_2 - 0)$$

$$\frac{1}{2} m v_B^2 = -mgh_2 + \frac{1}{2} m v_A^2$$

$$v_B = \sqrt{-2gh_2 + v_A^2}$$

$$= \sqrt{-2(9.8 \text{ m/s}^2)(8 \text{ m}) + (15.3 \text{ m/s})^2}$$

$$v_B = 8.8 \text{ m/s} \quad \checkmark \text{ (or } 8.9 \text{ m/s)}$$

final point is B
 initial point is A

- (c) As he begins to descend down a straight, 18.0 m long slope, he slows his skateboard down by using friction on the tail of the board. He is able to produce a friction force with magnitude of 120.0 N. What is his speed when he reaches the bottom of the slope (point C)?

Terwyl hy afwaarts teen die reguit skuinsvlak (met lengte 18.0 m) beweeg, gebruik hy wrywing op die eindpunt van sy skaatsplank om stadiger te beweeg. Hy produseer 'n wrywingskrag van 120.0 N op die wyse. Wat is sy spoed wanneer hy die onderpunt van die skuinsvlak (punt C) bereik? (4)

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0 ; \quad W_{\text{ext}} = 0 \quad \text{isolated system}$$

$$K_f - K_i + U_f - U_i + f_{kd} = 0 \quad \text{initial is at point D}$$

$$\frac{1}{2}mv_C^2 - 0 + 0 - mgh_i + f_{kd} = 0 \quad \text{final point is at C}$$

$$f_k = 120 \text{ N}$$

$$v_C = \sqrt{2gh_i - \frac{2f_{kd}}{m}}$$

$$= \sqrt{(2)(9.8 \text{ m/s}^2)(12 \text{ m}) - \frac{2(120 \text{ N})(18 \text{ m})}{45 \text{ kg}} \times \frac{1 \text{ kg} \cdot \text{m/s}^2}{1 \text{ N}}}$$

$$v_C = 11.8 \text{ m/s} \quad \checkmark$$

4 Question 4 / Vraag 4

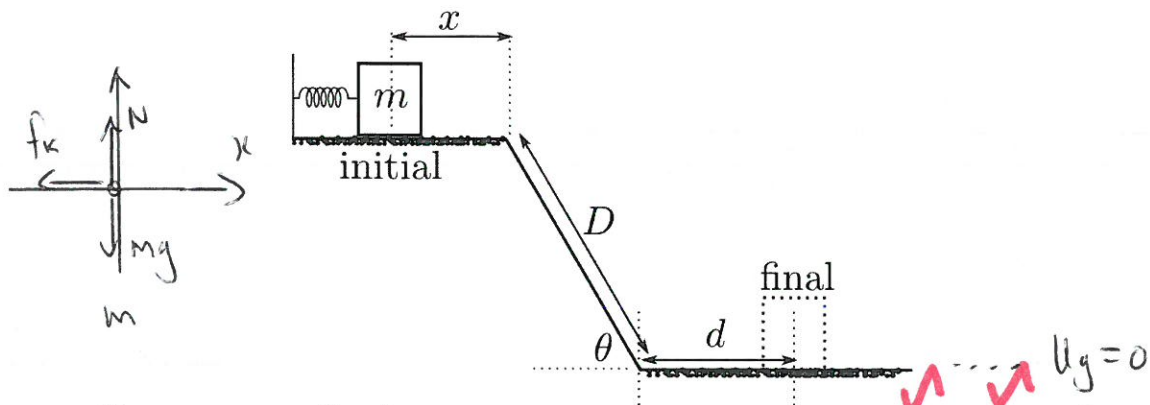
[8]

A small block of mass m is launched from rest by a horizontal spring as shown in the figure. The block slides on a track down a hill and comes to rest a distance d from the base of the hill. Kinetic friction is negligible on the hill, but the coefficient of kinetic friction between the block and the horizontal parts of the track is μ_k . The spring has a spring constant of k , and is compressed by a distance x (as shown in the figure) with the block attached. The block remains on the track at all times. Start with the conservation of energy equation $\Delta K + \Delta U + \Delta E_{\text{int}} = W_{\text{external forces}}$ and clearly explain your choice of initial and final values for each quantity in this equation that is applicable to the situation described and show that

$$d = \frac{D \sin \theta}{\mu_k} + \frac{1}{2} \frac{kx^2}{\mu_k mg} - x.$$

'n Klein blok met massa m word gelanseer vanuit rus deur 'n horisontale veer soos aangetoon in die figuur. Die blok gly op 'n baan teen 'n heuwel af, en kom tot rus, 'n afstand d van die basis van die heuwel. Kinetiese wrywing is weglaatbaar klein op die heuwel, maar die koëffisiënt van kinetiese wrywing tussen blok en die horisontale dele van die baan is μ_k . Die veer het 'n veerkonstante k , en word saamgepers met 'n afstand x (soos getoon in die figuur) met die blok daaraan vasgemaak. Die blok maak kontak met die baan ten alle tye. Begin met die behoud van energie vergelyking $\Delta K + \Delta U + \Delta E_{\text{int}} = W_{\text{external forces}}$ en verduidelik jou keuse van aanvangs- en finale waardes vir elke hoeveelheid in hierdie vergelyking soos vereis deur die probleem, en bewys dat

$$d = \frac{D \sin \theta}{\mu_k} + \frac{1}{2} \frac{kx^2}{\mu_k mg} - x.$$



$$\Delta K + \Delta U + \Delta E_{int} = W_{ext} ; W_{ext} = 0 \text{ isolated system}$$

$$K_f - K_i + U_{g,f} - U_{g,i} + U_{elas,f} - U_{elas,i} + \Delta E_{int} = 0$$

$$0 - 0 + 0 - mgD \sin \theta + 0 - \frac{1}{2} kx^2 + \Delta E_{int} = 0$$

$$\Rightarrow \Delta E_{int} = mgD \sin \theta + \frac{1}{2} kx^2 \quad - (1)$$

ΔE_{int} is the work done by friction

$$\Delta E_{int} = f_k l \quad \text{where } l \text{ is the total path length}$$

$$\Delta E_{int} = \mu_k N l = \mu_k mg l = \mu_k mg (x + d)$$

Eq ① becomes: $\mu_k mg x + \mu_k mg d = mgD \sin \theta + \frac{1}{2} kx^2$

$$\mu_k mg d = mgD \sin \theta + \frac{1}{2} kx^2 - \mu_k mg x$$

$$d = \frac{D \sin \theta}{\mu_k} + \frac{1}{2} \frac{k}{\mu_k mg} x^2 - x$$

5 Question 5 / Vraag 5

[6]

A single object is displaced by a net force $\Sigma \vec{F}$ acting only in the x -direction. We know from Newton's second law that the object moves with an acceleration $\vec{a}$. If the object moves through a displacement $\Delta \vec{r} = \Delta x \hat{i}$, the net work done on the object by the net force is

$$W_{net} = \int_{x_i}^{x_f} \Sigma F dx.$$

Use this result to prove the work-kinetic energy theorem that is $W_{net} = \Delta K$, where K is the kinetic energy of the object.

'n Enkele voorwerp word verplaas deur 'n netto krag $\Sigma \vec{F}$ wat slegs in die x -rigting uitgeoefen word. Ons weet vanaf Newton se tweede wet dat die voorwerp beweeg met 'n versnelling $\vec{a}$. Indien die voorwerp beweeg deur 'n verplasing $\Delta \vec{r} = \Delta x \hat{i}$, dan is die netto arbeid verrig deur die netto krag,

$$W_{net} = \int_{x_i}^{x_f} \Sigma F dx.$$

Gebruik hierdie resultaat om die arbeid-kinetiese energie teorie te bewys dat $W_{net} = \Delta K$, waar K die kinetiese energie van die voorwerp is.

We assume that the force is $\Sigma F = F_x(x)$ (varying force)

The net force causes an acceleration a_x .

Therefore the object has speed v_{ix} at x_i and speed v_{fx} at x_f ($v_f \neq v_i$)

$$W_{\text{net}} = \int_{x_i}^{x_f} F_x dx = \int_{x_i}^{x_f} m a_x dx$$

$$= m \int_{x_i}^{x_f} \frac{dv_x}{dt} dx \quad a_x = \frac{dv_x}{dt}$$

$$= m \int_{x_i}^{x_f} \frac{dv_x}{dx} \frac{dx}{dt} dx \quad \text{Chain rule}$$

$$= m \int_{x_i}^{x_f} v_x \frac{dv_x}{dx} dx \quad v_x = \frac{dx}{dt}$$

$$= m \int_{v_i}^{v_f} v_x dv_x$$

$$= m \left[\frac{1}{2} v_x^2 \right]_{v_i}^{v_f}$$

$$= m \left[\frac{1}{2} v_{fx}^2 - \frac{1}{2} v_{ix}^2 \right]$$

$$= \frac{1}{2} m v_{fx}^2 - \frac{1}{2} m v_{ix}^2$$

$$W_{\text{net}} = K_f - K_i = \Delta K$$

$$\text{where } K = \frac{1}{2} m v^2 \text{ is the KE.}$$

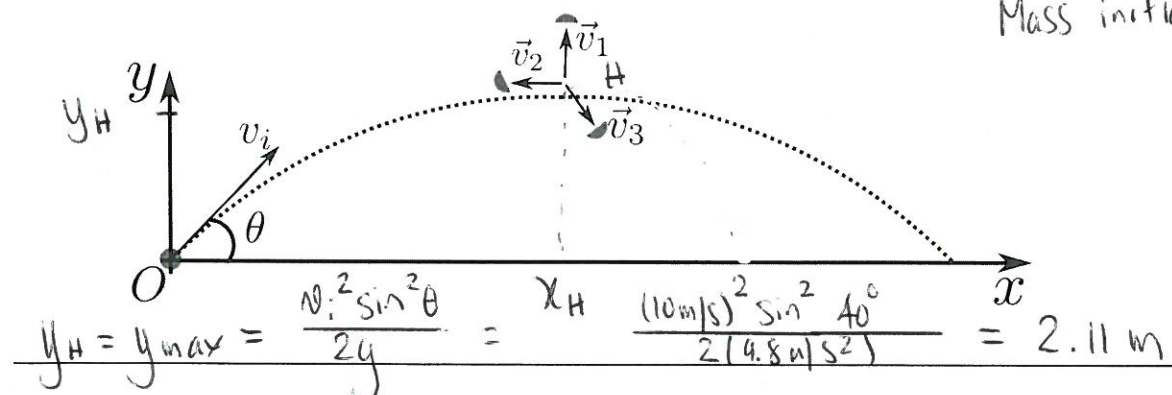
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Question 6 / Vraag 6

[11]

After several large firecrackers have been inserted into a bowling ball's finger holes, it is projected into the air using a homemade launcher and explodes in midair. During the launch, the ball is shot into the air with an initial speed of $v_i = 10.0 \text{ m/s}$ at an angle, $\theta = 40.0^\circ$. It explodes at the peak of its trajectory, breaking into three pieces of equal mass. One piece travels straight up with a speed of $v_1 = 3.0 \text{ m/s}$. Another piece travels straight back with a speed of $v_2 = 2.0 \text{ m/s}$. How long does it take for the third piece to hit the ground? What is the horizontal distance, as measured from the launch point, where the third piece hits the ground?

Nadat groot klappers in die vingergate van 'n kegelbal geplaas is, word dit met 'n tuisgemaakte lanseerder in die lug in geskiet, en dit ontplof in die lug. Gedurende die lansering, word die bal geskiet met 'n aanvangspoed van $v_i = 10.0 \text{ m/s}$ teen 'n hoek van $\theta = 40.0^\circ$ bo die horisontaal. Die bal ontplof by die bopunt van die baan van die bal, en breek in drie dele met gelyke massa. Een stuk beweeg reguit-op in die lug met 'n spoed van $v_1 = 3.0 \text{ m/s}$. 'n Ander stuk beweeg reguit-terugwaarts met 'n spoed van $v_2 = 2.0 \text{ m/s}$. Hoe lank neem dit die derde stuk om die grond te tref? Wat is die horisontale afstand, gemeet vanaf die lanseringspunt, waar die derde stuk die grond tref?



$$x_H = \frac{1}{2} R = \frac{1}{2} \left(\frac{v_i^2 \sin 2\theta}{g} \right) = \frac{1}{2} \left(\frac{(10 \text{ m/s})^2 \sin 80^\circ}{9.8 \text{ m/s}^2} \right) = (10.05 \text{ m})^{\frac{1}{2}} = 5.03 \text{ m}$$

At H : $v_{yH} = 0$ and $v_{xH} = v_{xi} = v_i \cos \theta$

Momentum conservation at point H where the ball explodes:

x-components: $p_{xi} = p_{xf}$

$$3m v_i \cos \theta = m v_{1x} + m v_{2x} + m v_{3x}$$

$$3m v_i \cos \theta = 0 - m v_2 + m v_{3x}$$

$$v_{3x} = 3v_i \cos \theta + v_2$$

$$v_{3x} = 3(10 \text{ m/s}) \cos 40^\circ + 2.0 \text{ m/s} = 24.98 \text{ m/s}$$

y-components: $p_{yi} = p_{yf}$

$$3m v_{yH} = m v_{1y} + m v_{2y} + m v_{3y}$$

$$0 = m v_1 + 0 + m v_{3y}$$

$$\Rightarrow v_{3y} = -v_1 = -3.0 \text{ m/s}$$

Piece no. 3 is launched with $v_{3x} = 24.98 \text{ m/s}$ and $v_{3y} = -3.0 \text{ m/s}$ and hits at point G:

$a_x = 0$

$$x_G = x_i + v_{3x} t_G \quad \text{and}$$

$$x_G = x_H + v_{3x} t_G$$

$$x_G = 5.03 \text{ m} + (24.98 \text{ m/s})(0.42 \text{ s})$$

$$x_G = 15.52 \text{ m}$$

$$y_G = y_i + v_{3y} t_G - \frac{1}{2} g t_G^2$$

$$0 = y_{\max} + v_{3y} t_G - \frac{1}{2} g t_G^2$$

$$t_G = \frac{-v_{3y} \pm \sqrt{v_{3y}^2 - 4(-\frac{1}{2}g)y_{\max}}}{-g}$$

$$t_G = -1.03 \text{ s} \quad \text{and} \quad t_G = 0.42 \text{ s}$$

7 Question 7 / Vraag 7

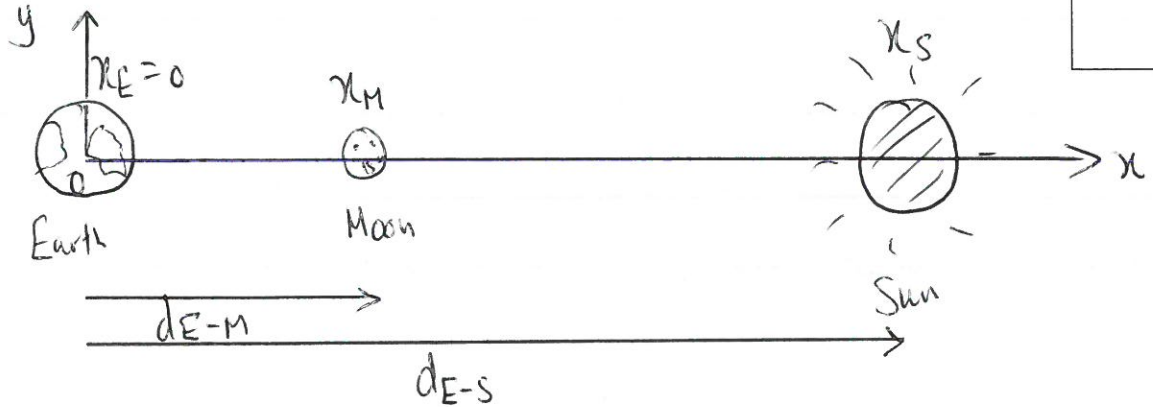
[9]

Consider the Earth, Moon, and Sun as a system of three objects in the following questions. The masses are $M_{\text{Sun}} = 1.989 \times 10^{30} \text{ kg}$, $M_{\text{Earth}} = 5.97 \times 10^{24} \text{ kg}$, and $M_{\text{Moon}} = 7.35 \times 10^{22} \text{ kg}$. The orbital distance between the Sun and the Earth is $d_{\text{S-E}} = 1.496 \times 10^{11} \text{ m}$, and the orbital distance between the Earth and the Moon is $d_{\text{E-M}} = 3.85 \times 10^8 \text{ m}$. Make a clear sketch (showing an appropriate coordinate system choice and correctly labelled) in each case to support your calculation.

Beskou die Aarde, Maan en Son as 'n stelsel van drie voorwerpe in die volgende vrae. Die massas is $M_{\text{Son}} = 1.989 \times 10^{30} \text{ kg}$, $M_{\text{Aarde}} = 5.97 \times 10^{24} \text{ kg}$, en $M_{\text{Maan}} = 7.35 \times 10^{22} \text{ kg}$. Die wentelbaan afstand tussen die Son en die Aarde is $d_{\text{S-E}} = 1.496 \times 10^{11} \text{ m}$, en die wentelbaan-afstand tussen die Aarde en die maan is $d_{\text{E-M}} = 3.85 \times 10^8 \text{ m}$. Maak 'n netjiese skets (toon jou koördinaat-stelsel en gee korrekte etikette) in elke geval om jou antwoord te ondersteun.

- (a) When there is a solar eclipse, the Moon is between the Earth and the Sun. How far is the center of mass of this three-particle system located away from the Earth?

Wanneer daar 'n sonsverduistering plaasvind, is die Maan tussen die Aarde en die Son. Hoe ver is die massa-middelpunt van die drie-deeltjie stelsel geleë vanaf die Aarde? (3)



$$x_{cm} = \frac{1}{M} (M_E x_E + M_M x_M + M_S x_S)$$

$$M = M_E + M_M + m_S = (5.97 \times 10^{24} \text{ kg} + 7.35 \times 10^{22} \text{ kg} + 1.989 \times 10^{30} \text{ kg})$$

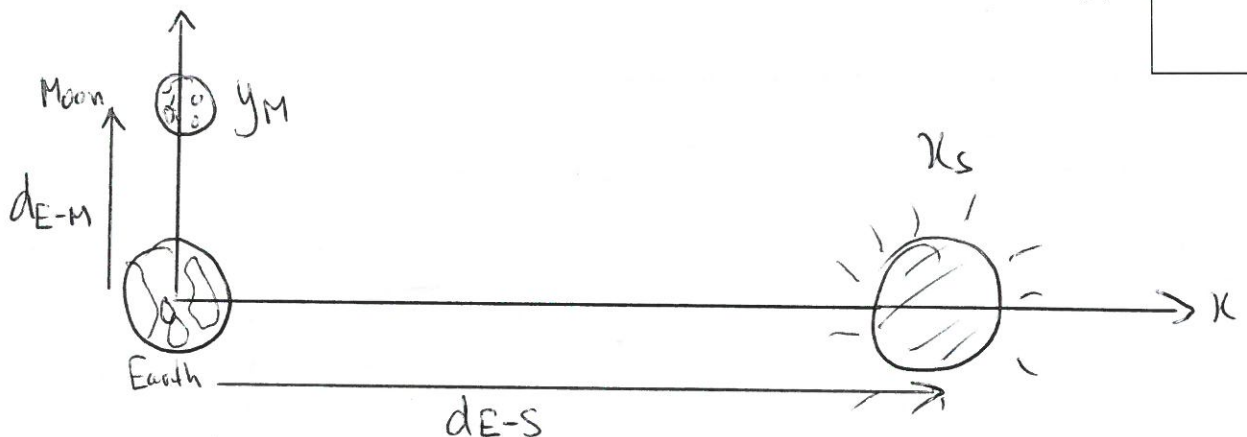
$$M = 1.989 \times 10^{30} \text{ kg} \quad \checkmark$$

$$x_{cm} = \frac{1}{1.989 \times 10^{30} \text{ kg}} [M_E (0) + (7.35 \times 10^{22} \text{ kg})(3.85 \times 10^8 \text{ m}) + (1.989 \times 10^{30} \text{ kg})(1.496 \times 10^{11} \text{ m})] \quad \checkmark$$

$$x_{cm} = 1.49598 \times 10^{11} \text{ m} \quad \checkmark$$

- (b) When the Moon's phase is a "quarter Moon", the Earth, the Moon, and the Sun are at the vertices (corners) of a right triangle. Assume all the orbits are circular. How far is the center of mass of the Earth-Moon-Sun system located away from Earth during this phase of the Moon?

Wanneer die Maan se fase 'n "kwart-Maan" is, dan is die Aarde, die Maan en die Son by die hoekpunte van 'n regte-hoek-driehoek. Neem aan al die wentelbane is sirkelvormig. Hoe ver is die massa-middelpunt van die drie-deeltjie stelsel geleë vanaf die Aarde gedurende hierdie fase van die Maan? (6)



$$x_{cm} = \frac{1}{M} [M_E x_E + M_M x_M + M_S x_S]$$

$$= \frac{1}{1.989 \times 10^{30} \text{ kg}} [M_E (0) + M_M (0) + (1.989 \times 10^{30} \text{ kg})(1.496 \times 10^{11} \text{ m})]$$

$$x_{cm} = 1.496 \times 10^4 \text{ m} \quad \checkmark$$

$$y_{cm} = \frac{1}{M} [M_E y_E + M_M y_M + M_S y_S]$$

$$= \frac{1}{1.989 \times 10^{30} \text{ kg}} [M_E(0) + (7.35 \times 10^{22} \text{ kg})(3.85 \times 10^8 \text{ m}) + M_S(0)]$$

$$y_{cm} = 14.2270 \text{ m} \quad \checkmark$$

$$\text{Distance of com from earth is } \sqrt{x_{cm}^2 + y_{cm}^2} = 1.496 \times 10^4 \text{ m} \quad \checkmark$$

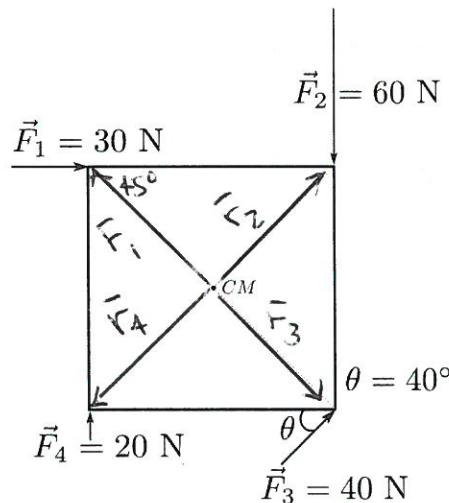
8 Question 8 / Vraag 8

[7]

A square plate with side lengths of 2.0 m each can rotate around an axle passing through its center of mass (CM) and perpendicular to its face. There are four forces acting on the plate at different points as shown in the figure. What is the net torque acting on the plate?

'n Vierkantige plaat met sy lengtes van 2.0 m kan roteer om 'n as wat deur die massa-middelpunt (CM) loop, loodreg t.o.v. die aansigkant. Vier kragte word toegepas op die plaat op verskillende punte soos aangetoon in die figuur. Wat is die netto wringkrag wat op die plaat inwerk?

Each torque points in either $\hat{k}$ or $-\hat{k}$ direction since the forces and $\vec{r}$'s are in the x - y -plane.



$$|\vec{r}_1| = |\vec{r}_2| = |\vec{r}_3| = |\vec{r}_4| = |\vec{r}|$$

$$= \frac{\sqrt{(2\text{ m})^2 + (2\text{ m})^2}}{2}$$

$$r = \frac{\sqrt{8}}{2} \text{ m} \quad \checkmark$$

CW rotation

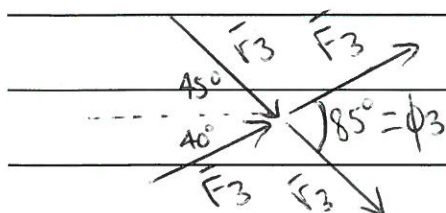
$$\tau_1 = -F_1 r_1 \sin \phi_1 = -(30 \text{ N}) \left(\frac{\sqrt{8}}{2} \text{ m} \right) \sin 135^\circ = -30 \text{ N}\cdot\text{m} \quad \checkmark$$

$$\tau_2 = -F_2 r_2 \sin \phi_2 = -(60 \text{ N}) \left(\frac{\sqrt{8}}{2} \text{ m} \right) \sin 135^\circ = -60 \text{ N}\cdot\text{m} \quad \checkmark$$

$$\tau_4 = -F_4 r_4 \sin \phi_4 = -(20 \text{ N}) \left(\frac{\sqrt{8}}{2} \text{ m} \right) \sin 135^\circ = -20 \text{ N}\cdot\text{m} \quad \checkmark$$

CCW rotation

$$\tau_3 = +F_3 r_3 \sin \phi_3 = +(40 \text{ N}) \left(\frac{\sqrt{8}}{2} \text{ m} \right) \sin 85^\circ = 56.35 \text{ N}\cdot\text{m} \quad \checkmark$$



$$\tau_{net} = \tau_1 + \tau_2 + \tau_3 + \tau_4$$

$$\tau_{net} = -53.65 \text{ N}\cdot\text{m} \quad \checkmark$$

$$\tau_{net} = (-53.65 \hat{k}) \text{ N}\cdot\text{m} \quad \text{vector!} \quad \checkmark$$

9 Question 9 / Vraag 9

[7]

The flywheel of a steam engine starts to rotate from rest with a constant angular acceleration of $\alpha = 1.43 \text{ rad/s}^2$. The flywheel undergoes this constant angular acceleration for $t = 25.9 \text{ s}$ and then continues to rotate at a constant angular velocity, ω . After the flywheel has been rotating for 59.5 s , what is the total angle through which it has rotated since it started?

Die vliegwiel van 'n stoomenjin begin om te roteer met 'n konstante hoekversnelling van $\alpha = 1.43 \text{ rad/s}^2$. Die vliegwiel besit hierdie hoekversnelling vir $t = 25.9 \text{ s}$ waarna dit verder roteer teen 'n konstante hoeksnelheid, ω . Nadat die vliegwiel vir 59.5 s geroteer het, wat is die totale hoek waardeur dit geroteer het vandat die rotasie begin het?

The flywheel starts at $\theta_i = 0$ with an angular speed $\omega_i = 0$ and accelerates for a time t_a with constant angular acceleration α .

$$\Rightarrow \theta_a = \theta_i + \omega_i t_a + \frac{1}{2} \alpha t_a^2$$

$$\theta_a = 0 + 0 + \frac{1}{2} \alpha t_a^2 = \frac{1}{2} \alpha t_a^2 \text{ is the angular displacement during } t_a.$$

The total time of rotation is t_b . For a time $(t_b - t_a)$ it rotates through an angle $\theta_b = \omega(t_b - t_a)$ but $\omega = \alpha t$, the final angular speed reached after time t_a

$$\Rightarrow \theta_b = \omega(t_b - t_a) = \alpha t_a(t_b - t_a)$$

$$\theta_{\text{total}} = \theta_a + \theta_b$$

$$= \frac{1}{2} \alpha t_a^2 + \alpha t_a(t_b - t_a)$$

$$= \frac{1}{2} (1.43 \text{ rad/s}^2)(25.9 \text{ s})^2 + (1.43 \text{ rad/s}^2)(25.9 \text{ s})[59.5 \text{ s} - 25.9 \text{ s}]$$

$$\theta_{\text{total}} = +1724.07 \text{ rad}$$

END OF TEST / EINDE VAN TOETS